



Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

Telecom Policy: TE 452

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Course Objective

- ❖ To introduce students to the principles of telecommunications regulations.



Course Content

❖ Background to regulation

- The economics of markets
- Privatization, Liberalization, Regulation

❖ Overview of Telecommunications Regulation

- Regulatory objectives
- Regulatory organisations
- The regulatory process
- Principles of effective regulation

❖ Framework for regulation

- Legal frameworks Styles of regulation
- Enforcement
- Funding and accountability
- Dangers of regulation (regulatory paralysis, regulatory capture, etc.)



Course Content

❖ Licensing Telecommunications Services

- Scarce Resources in Telecommunications (Spectrum and Numbers)
- Introduction to licensing
- Types of licensing regimes
- The licensing process
- Licensing practices
- Contents of licensing
- Other regulatory instruments



Course Content

❖ Interconnection

- Interconnection principles
- Interconnection procedures
- Financial terms of interconnection
- Technical and operational conditions



Course Content

❖ Price Regulation

- Introduction
- Approaches
- Price cap regulation
- Price cap variation
- Rate rebalancing and price caps
- International accounting



Course Content

❖ Competition Policy

- General principles
- Basic concepts of competition policy
- Remedies for anti-competitive conduct
- Mergers, acquisitions and other corporate combinations



Course Content

❖ Universal Access and Service (UAS)

- Introduction
- Funding UAS

❖ Current issues in regulation

- Regulation in the era of technological convergence
- Regulating over-the-top (OTT) services
- Other trending topics related to regulation

❖ Case studies of national telecommunications policy documents

❖ Case studies of regulatory strategies from other countries



Learning Outcomes

- ❖ At the end of this course, students should be able to;
 - Understand the principles of telecommunications regulation
 - Explain the importance of regulation
 - Know current trends in telecommunications regulation
 - Contribute to discussions related to telecommunications policy and regulation



Learning Activities

- ❖ Two hours of lectures weekly
- ❖ Group presentations on topics related to telecommunications regulation
- ❖ Online forums and discussions*



Classroom Ethics

- ❖ No cell phones*
- ❖ No eating
- ❖ No intimidation
- ❖ No lateness



Reading List

- ❖ Telecommunications Policy, John Buckley, IET
- ❖ Telecommunications Regulation Handbook, Hank Intven and McCarthy Tetrault (editors), infoDev
- ❖ WTO Reference Paper on Regulation
- ❖ ICT Regulation Toolkit, infoDev and ITU,
- ❖ [http:// www.ictregulationtoolkit.org](http://www.ictregulationtoolkit.org)



Assessment

- ❖ Quizzes ----- 5 marks
- ❖ Presentations ----- 10 marks
- ❖ Mid-Sem ----- 15 marks
- ❖ Exams ----- 70 marks



Background to Regulations

- ❖ The economics of markets
- ❖ Privatization, Liberalization, Regulation



The Economics of Markets

- ❖ What is a **market**?
- ❖ A market is a group of buyers and sellers of a particular good or service.
- ❖ The buyers as a group determine the demand for the product, and the sellers as a group determine the supply of the product.
- ❖ Some markets are highly organized, such as in ***auctions***, where buyers and sellers meet at a specific time and place where an auctioneer helps set prices and arrange sales.



The Economics of Markets

- ❖ More often, markets are less organized. For example, consider the market for ice cream in a particular town.
 - **Discussion (Group):** Which other forms of market do you know?
 - (Non physical, such as online stores)
- ❖ Are they organized or not?



Principles of the Market Economy

- ❖ **Market economies** are founded on the principle that:
- ❖ economic efficiency and personal well being will be at their highest when individuals are free to follow their own interests as buyers and sellers, and
- ❖ where transactions are enacted between free and willing parties.
- ❖ The economy thus functions as a complex system of independently interacting agents.



Principles of the Market Economy

Such a system can be self-optimizing in that:

- ❖ it produces a stable equilibrium of supply, consumption and prices that is optimal.

Common alternatives to market economies are

- ❖ **traditional economies** and
- ❖ **command economies.**



Traditional Economy

- ❖ Traditional economies are
 - typical of tribal societies
 - economic affairs follow a customary pattern
 - local leaders have the ultimate say in matters of allocation
 - works well in static societies
 - may fail to respond to innovation and a changing environment.



Command Economy

- ❖ In a command economy a powerful agent (usually the government) determines what people will produce, earn and consume.
- ❖ Twentieth century experience shows that this model fails because of
 - the intractable complexity of the control task
 - vulnerability to managerial corruption.
- **Discussion:** Between traditional and command economies, which one would you prefer? Why? (**Group**)



Mixed Economies

- ❖ Mixed economies are tools that work well to achieve desired outcomes.
- ❖ They are morally neutral in that
 - it is people, acting both as individuals and through elected governments, that choose the outcomes they want.
 - Markets can succeed effectively



Mixed Economies

- ❖ Markets may also fail
- ❖ In the event of failure
 - Governments (through regulators and other institutions) intervene to correct it.
 - Command measures, blended with market economies create the mixed economies



Demand

- ❖ The **quantity demanded** of any good is the amount of the good that buyers are willing and able to purchase.
- ❖ Many things determine the quantity demanded of any good.
- ❖ One determinant plays a central role—the price of the good.

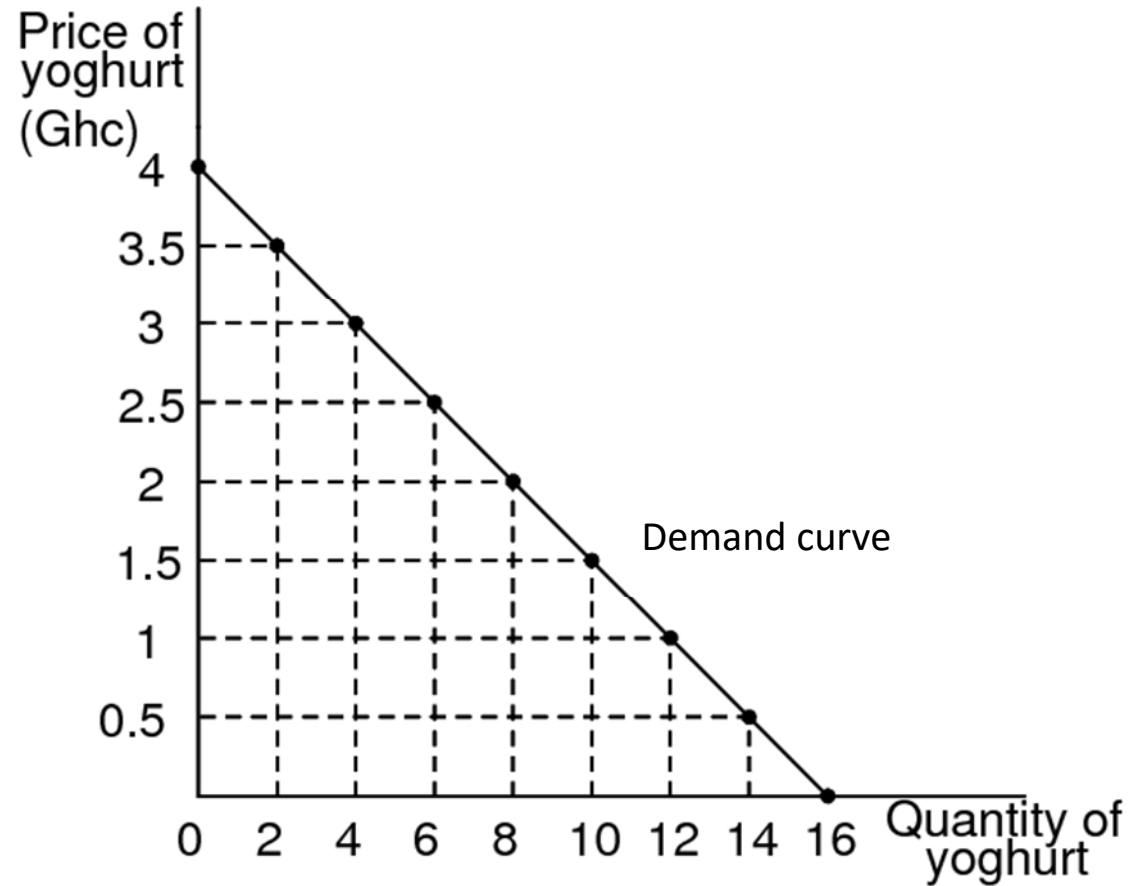


Demand

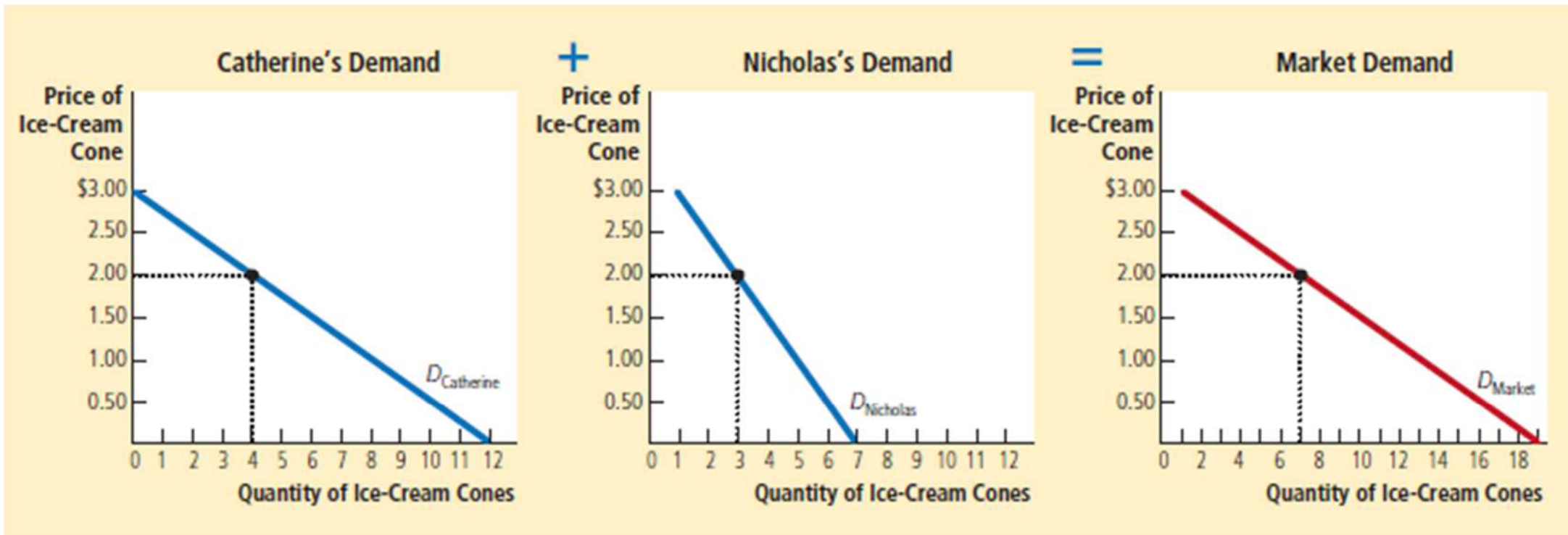
- ❖ If the price of fresh yoghurt rises to Gh50 per bottle, you would buy less fresh yoghurt. You might buy fanyogo or fanice instead.
 - ❖ If the price of fresh yoghurt falls to Gh0.50 per bottle, you would buy more.
- ❑ **Law of Demand:** Other things being equal, when the price of a good rises, the quantity demanded of the good falls, and when the price falls, the quantity demanded rises.



Demand



Market Demand

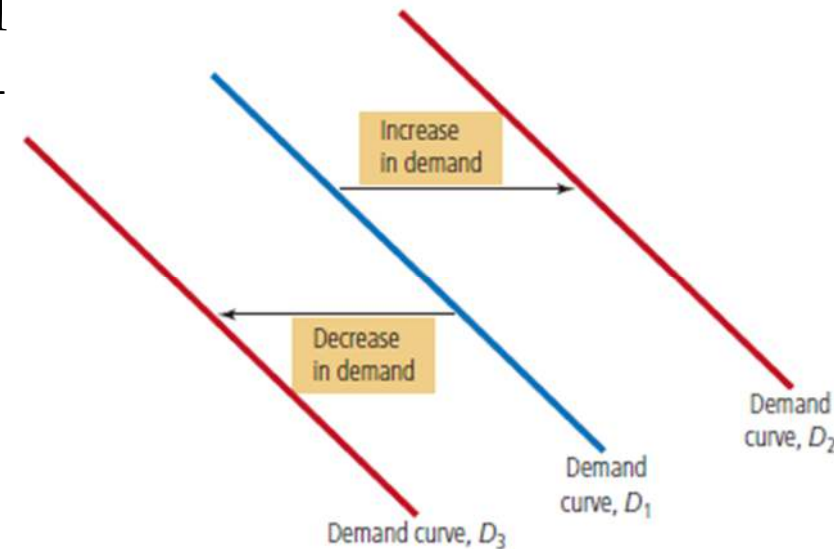


Demand Shift

- ❖ If something happens to alter the quantity demanded at any given price, the demand curve shifts.
- ❖ For example, suppose the Ghana Medical Association discovers that tiger nuts can cure covid-19.

Many variables that can shift the demand curve.

- ❖ Income
- ❖ Prices of related goods
- ❖ Taste
- ❖ Expectations
- ❖ Number of buyers

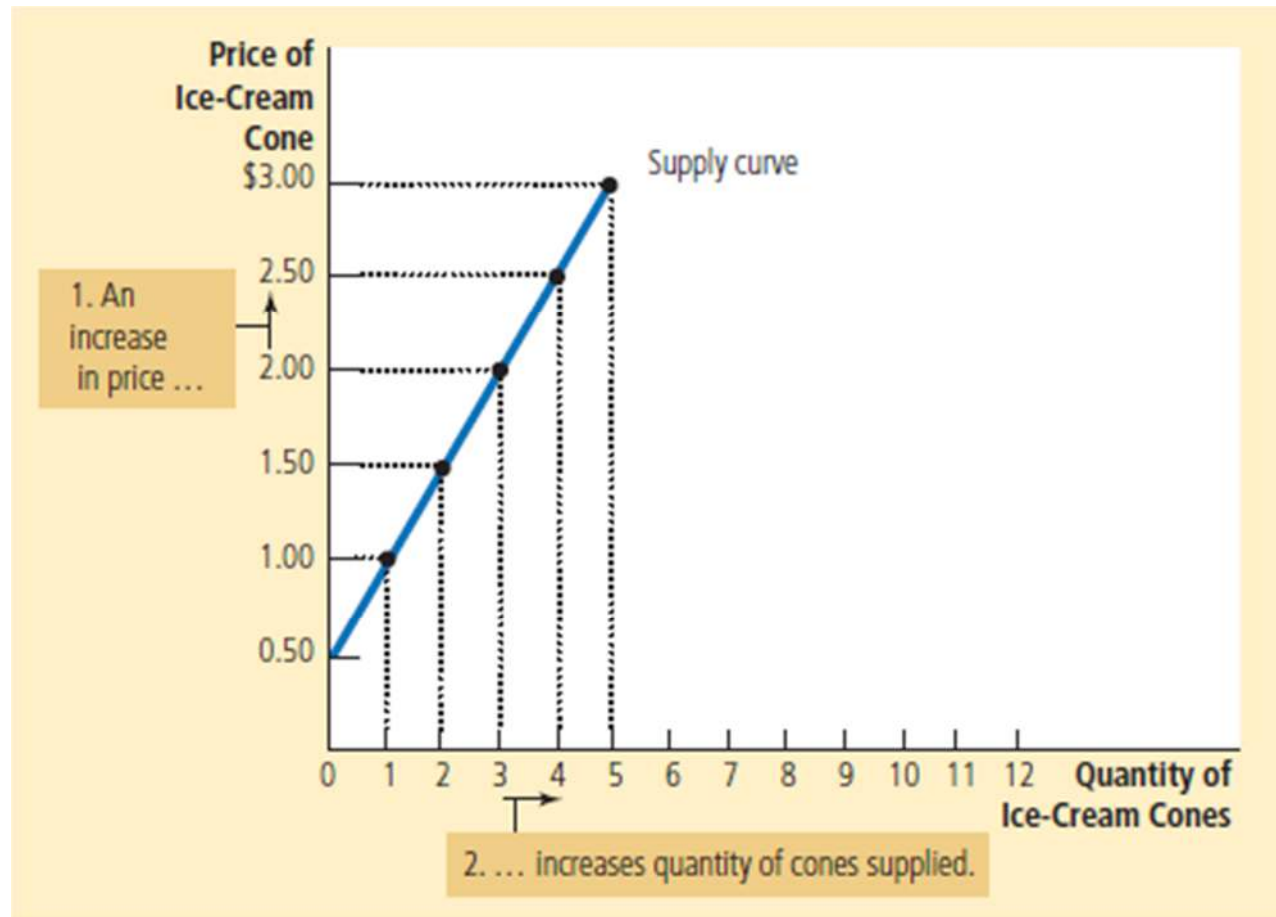


Supply

- ❖ The **quantity supplied** of any good or service is the amount that sellers are willing and able to sell.
 - ❖ When the price of ice cream is high, selling ice cream is profitable, and so the quantity supplied is large.
 - ❖ When the price of ice cream is low, the business is less profitable, so sellers produce less ice cream.
- ❑ **Law of Supply:** Other things being equal, when the price of a good rises, the quantity supplied of the good also rises, and when the price falls, the quantity supplied falls as well.

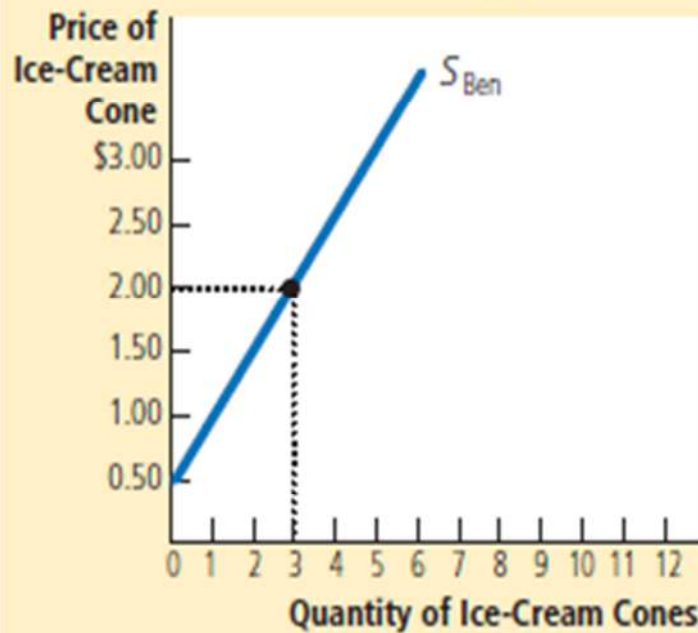


Supply



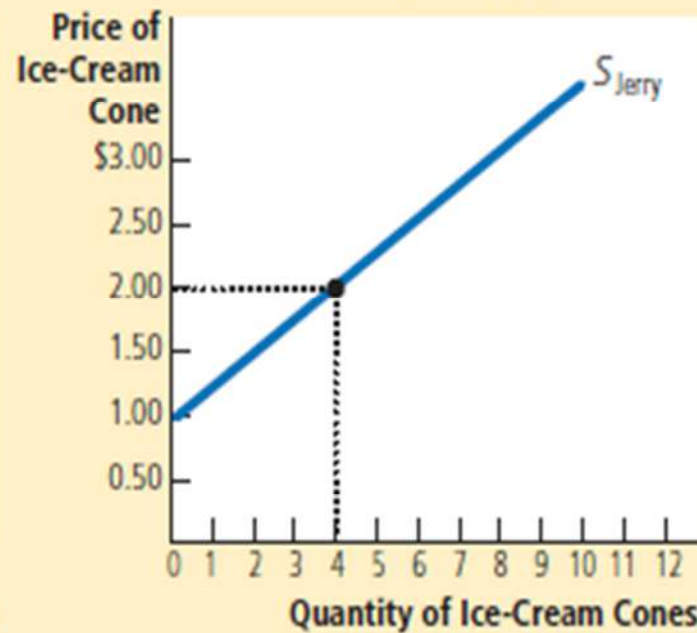
Market Supply

Ben's Supply



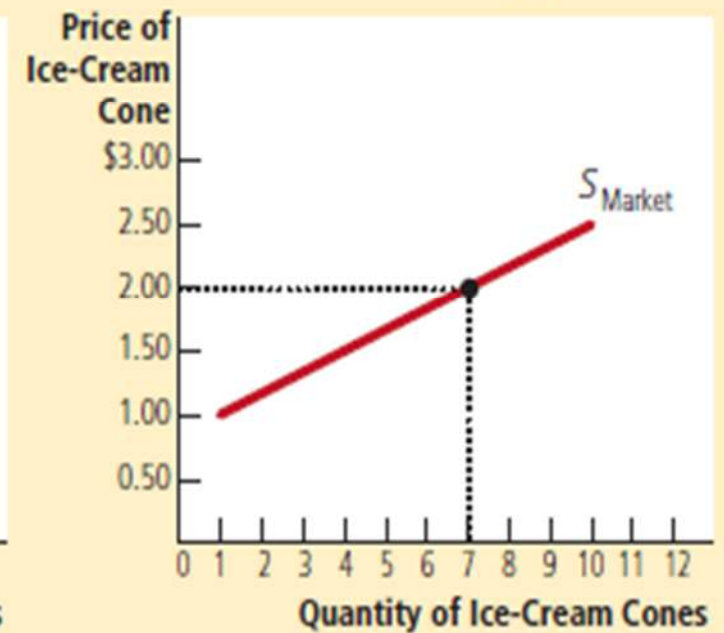
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Jerry's Supply



=

Market Supply



Supply Shift

❖ Because the market supply curve is drawn holding other things constant, when one of these factors changes, the supply curve shifts.

❖ Suppose the price of sugar falls.

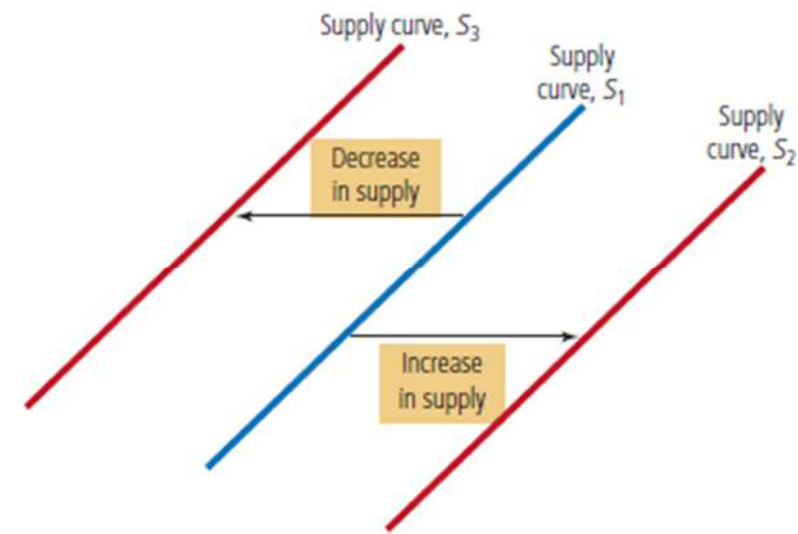
Many variables that can shift the supply curve.

❖ Input Prices

❖ Technology

❖ Expectations

❖ Number of Sellers



Elasticity of Supply and Demand

- Elasticity is a measure of how much buyers and sellers respond to changes in market conditions.

Demand elasticity:

- Demand for a good is said to be elastic if the quantity demanded responds substantially to changes in the price.
- Demand is said to be inelastic if the quantity demanded responds only slightly to changes in the price.



Elasticity of Supply and Demand

Supply elasticity:

- Supply of a good is said to be elastic if the quantity supplied responds substantially to changes in the price.
- Supply is said to be inelastic if the quantity supplied responds only slightly to changes in the price.



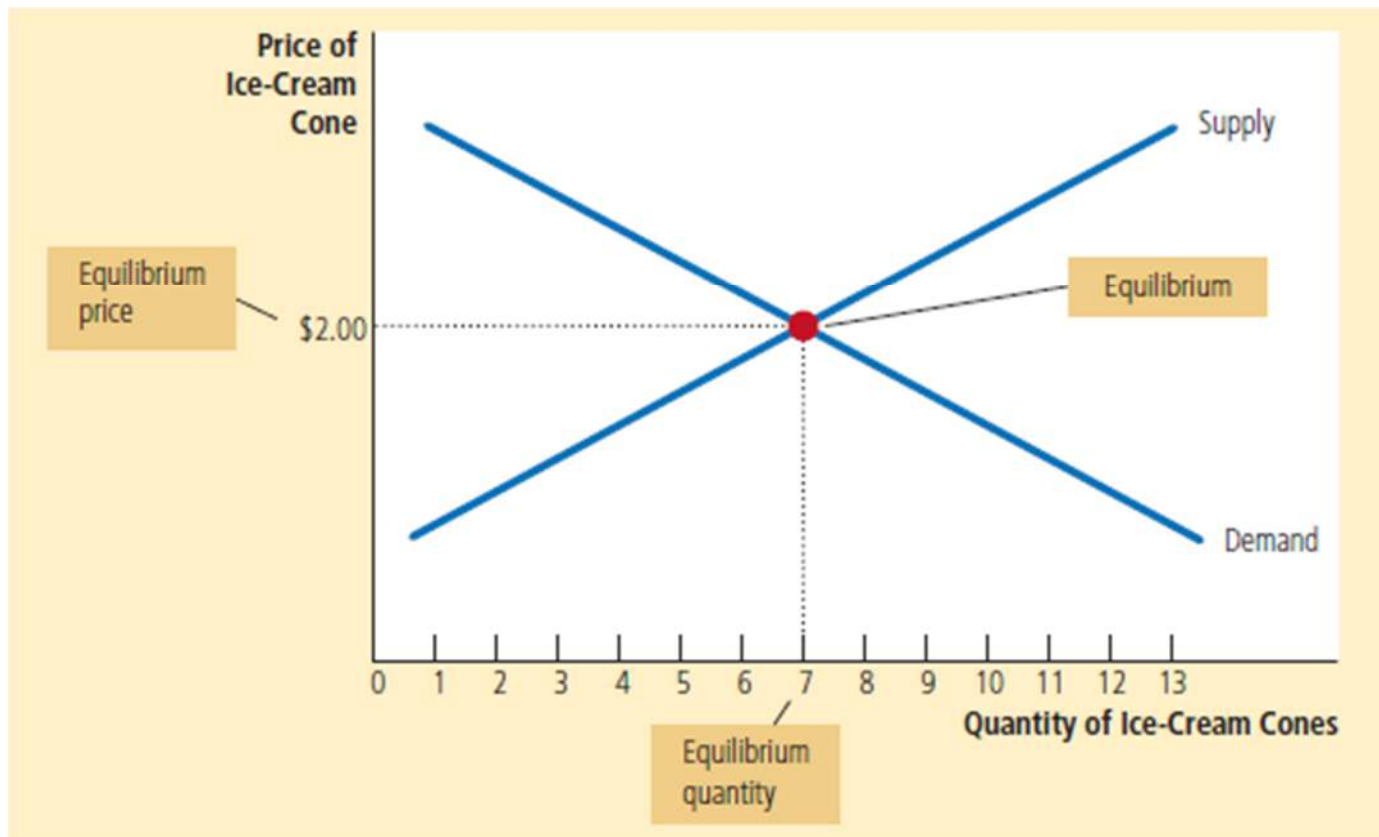
Equilibrium

- ❖ **Equilibrium** is the one point at which the supply and demand curves intersect.
- ❖ The price at this intersection is called the equilibrium price.
- ❖ The quantity is called the equilibrium quantity.
- ❖ At the equilibrium price, the quantity of the good that buyers are willing and able to buy exactly balances the quantity that sellers are willing and able to sell.
- ❖ The equilibrium price is sometimes called the *market-clearing price*.

Why? (Group)



Equilibrium



Impediments to Market Equilibrium

- To achieve some policy goals, governments can interfere with market forces that affect supply and demand by introducing
 - ❖ Price controls
 - ❖ Taxes
- These controls affect the market equilibrium, hence, the market price and quantity (supplied or demanded)



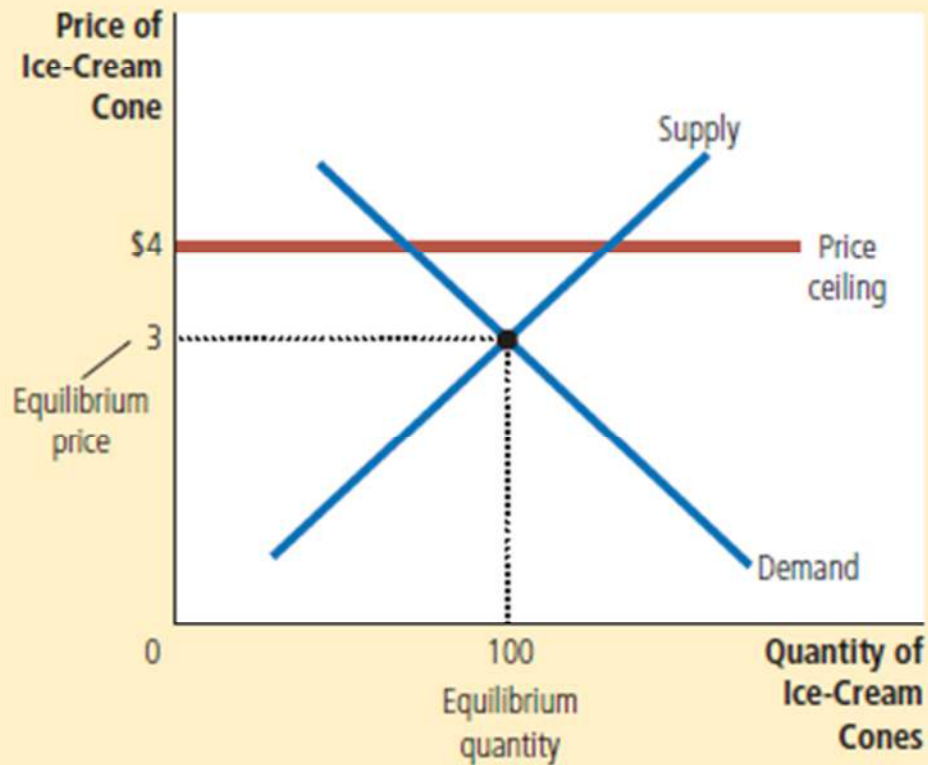
Price controls

- Price Ceilings and Floors
 - ❖ Sometimes the government intervenes in a market to ensure that the price stays below the market equilibrium price.
 - ❖ Prices below the market equilibrium normally rise therefore such policy interventions are called **price ceilings**, because they prevent the price from rising as high as it may have gone if left alone.
 - ❖ Prices hit the ceiling and then go no higher.
 - ❖ The opposite of market intervention is a **price floor**, by which the government keeps the price above its market equilibrium value.

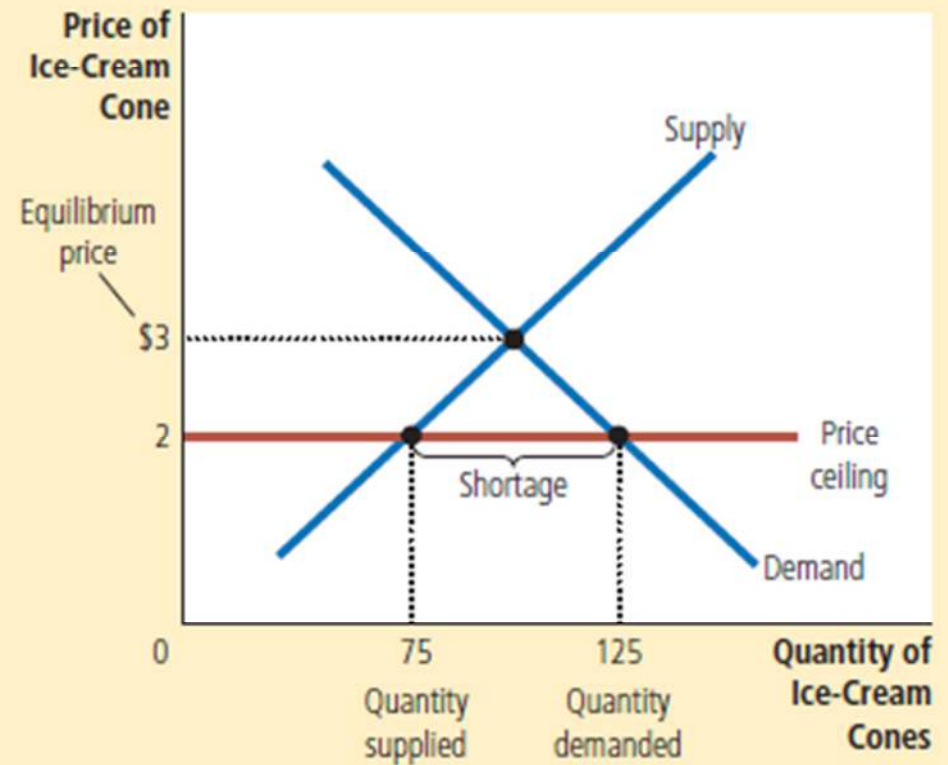


Price Ceilings and Floors

(a) A Price Ceiling That Is Not Binding

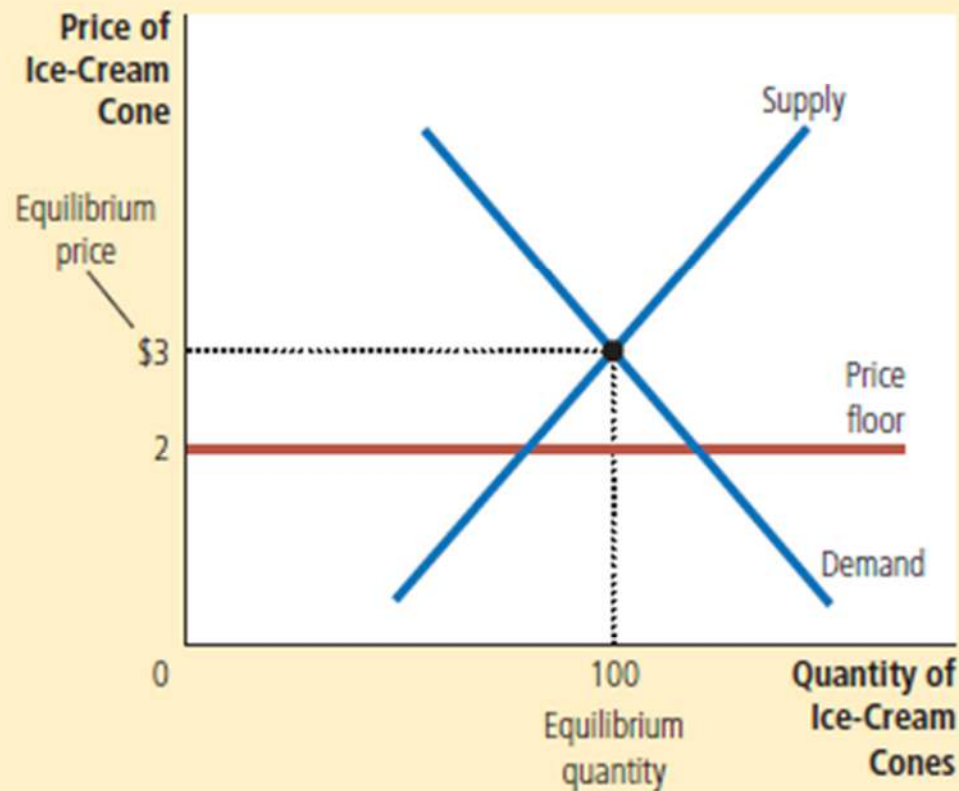


(b) A Price Ceiling That Is Binding

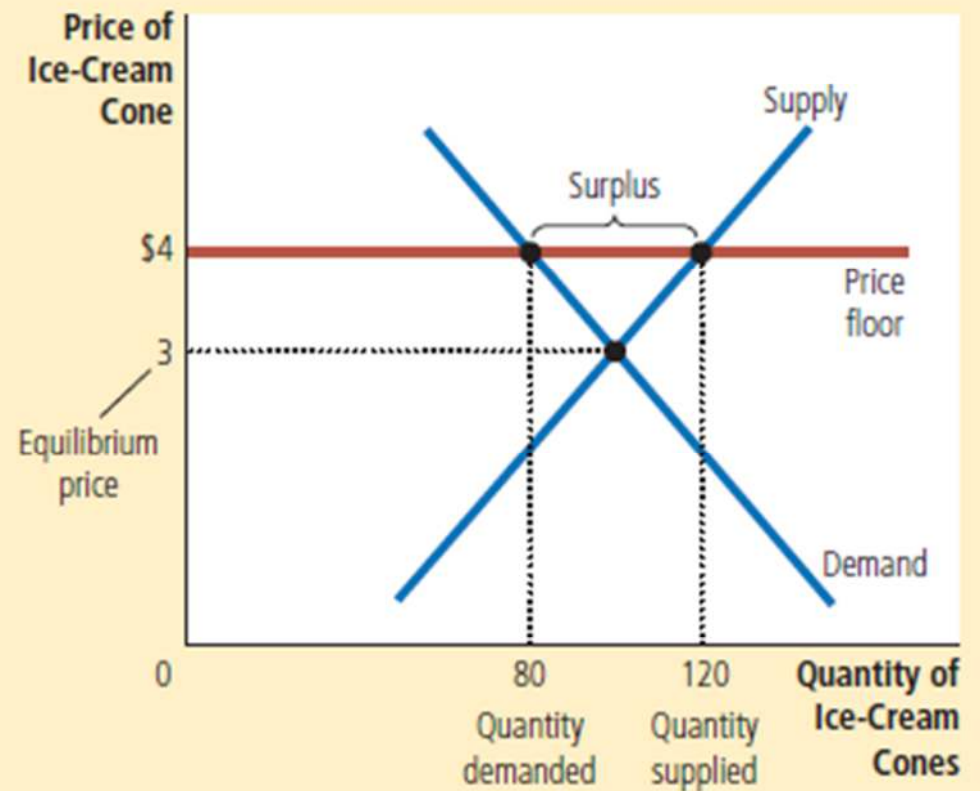


Price Ceilings and Floors

(a) A Price Floor That Is Not Binding



(b) A Price Floor That Is Binding



Taxes

- Taxes are introduced on
 - ❖ Sellers of goods
 - ❖ Buyers of goods
 - ❖ **Tax incidence** refers to how the burden of a tax is distributed among the various people who make up the economy.



How Taxes on Sellers Affect Market Outcomes

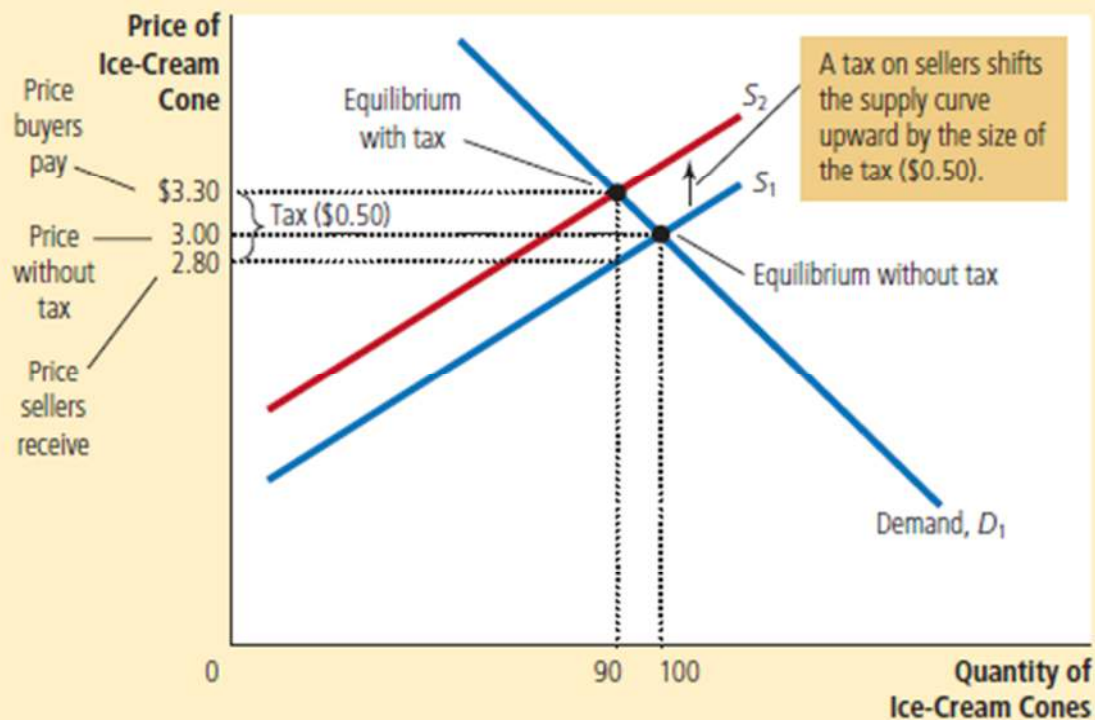


FIGURE 6

A Tax on Sellers

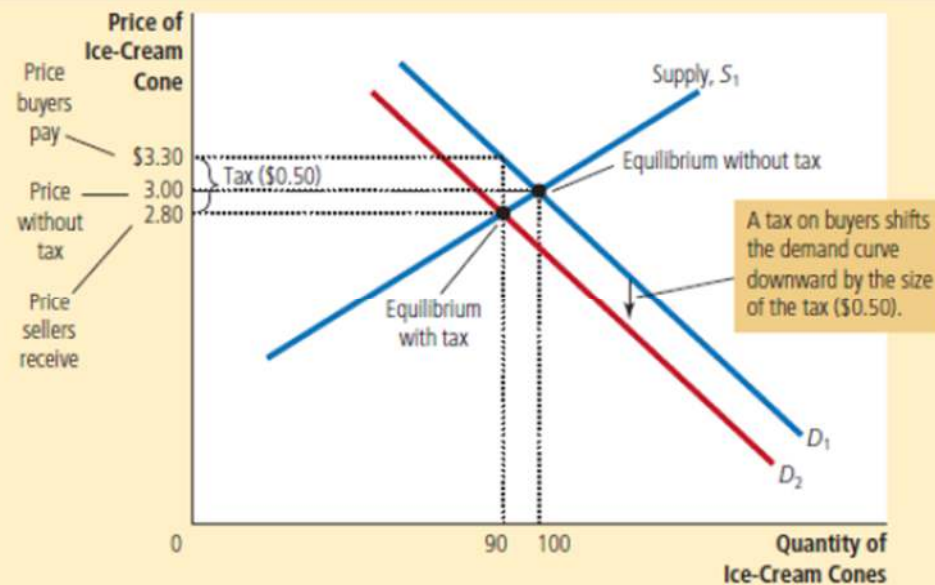
When a tax of \$0.50 is levied on sellers, the supply curve shifts up by \$0.50 from S_1 to S_2 . The equilibrium quantity falls from 100 to 90 cones. The price that buyers pay rises from \$3.00 to \$3.30. The price that sellers receive (after paying the tax) falls from \$3.00 to \$2.80. Even though the tax is levied on sellers, buyers and sellers share the burden of the tax.

How Taxes on Buyers Affect Market Outcomes

FIGURE 7

A Tax on Buyers

When a tax of \$0.50 is levied on buyers, the demand curve shifts down by \$0.50 from D_1 to D_2 . The equilibrium quantity falls from 100 to 90 cones. The price that sellers receive falls from \$3.00 to \$2.80. The price that buyers pay (including the tax) rises from \$3.00 to \$3.30. Even though the tax is levied on buyers, buyers and sellers share the burden of the tax.



Competitive Markets

- ❖ **Competitive market** is a market in which there are so many buyers and so many sellers that each has a negligible impact on the market price.
- We assume that markets are *perfectly competitive*. To achieve this
 - ❖ The goods offered for sale are all exactly the same
 - ❖ The buyers and sellers are so numerous that no single buyer or seller has any influence over the market price.



Monopoly

- ❖ A firm that has no competitors in its industry is called a monopoly.
- ❖ Governments typically go out of their way to break up monopolies and replace them with competitive industries that generate lower prices and higher output.
- ❖ Governments may also create monopolies intentionally in other situations.
- ❖ For example, governments issue patents, which give monopoly rights to inventors to sell and market their inventions.



Monopoly

- In an industry with only one monopoly firm rather than lots of small competitive firms, three socially harmful things occur:
 - ❖ The monopoly firm produces less output than firms in a competitive industry.
 - ❖ The monopoly firm sells its output at a higher price than if the industry was competitive.
 - ❖ The monopoly firm's output is produced less efficiently and at a higher cost than the output produced by firms in a competitive industry.



Monopoly

- ❖ An industry is a natural monopoly when a single firm can supply a good or service to an entire market at a lower cost than could two or more. (Example?) (Group)
- ❖ Where a natural monopoly exists, then it is not possible to obtain a competitive market, or at least not one that will generate allocative efficiency.
- ❖ In that case, a government can choose to live with the monopoly and possibly regulate it, or try to enforce wasteful competition.
- ❖ Examples of natural monopolies include, water supply, power distribution, telecommunications.



Telecommunication Sector Reforms

- **The Telecommunication Sector before the 1980s**
 - ❖ Telecoms services were provided by state-owned monopolies.
 - ❖ There were two main factors underlying this monopoly paradigm:
- **Economic:**
 - ❖ In the early days of telecommunications, the cost of building networks was very high and the level of demand for services was very low.
 - ❖ It would take a long time for any network to recover its costs or make a profit, making the potential market unviable for second operators.
 - ❖ The industry therefore exhibited natural monopoly characteristics.



Telecommunication Sector Reforms

- **Political:**
- Most governments at the time were keen to maintain state power over communications and, in practice, saw telecommunications as an extension of postal networks which were already within their control.
- ❖ Government departments were therefore responsible for both policy concerning telecommunications and for its implementation, for the management of networks and the delivery of services.



Telecommunication Sector Reforms

- ❖ Usually, these departments were responsible for postal services as well as telecommunications.
- **The Telecommunication Sector since the 1980s**
- ❖ From the early 80s, policies of governments have been towards the restructuring of the telecommunications sector through liberalization, privatization and regulation.



Telecommunication Sector Reforms

- **Liberalization:**
- This represents the process of transition from a monopoly market to a competitive one.
- In the case of telecommunications, liberalization has meant two things:
 - ❖ The removal of exclusive rights ("exclusivities") from the former monopoly fixed network operator, which is often called the "incumbent", so that in future it must offer what used to be monopoly services in competition with other businesses. (Allowing new entrants)



Telecommunication Sector Reforms

- ❖ The award of licenses for all new services including mobile networks and services, value-added services and the internet on a competitive basis.



Telecommunication Sector Reforms

- **Privatization:**
- This is the transfer of ownership of a business from the state to the private sector.
- In the case of telecommunications, privatization has meant two things:
 - ❖ The privatization, in whole or in part, of the telecommunications business(es) that were owned by government, either through a share sale (typical in industrial countries) or through sale of a stake in the company to a strategic investor (more common in developing countries).



Telecommunication Sector Reforms

- ❖ The award of licenses in all new services including mobile networks and services, value-added services and the internet to private companies.
- ❖ As with liberalization, as a result of the latter, even where the incumbent remains state-owned, the majority of the telecommunications sector is now usually in private ownership.



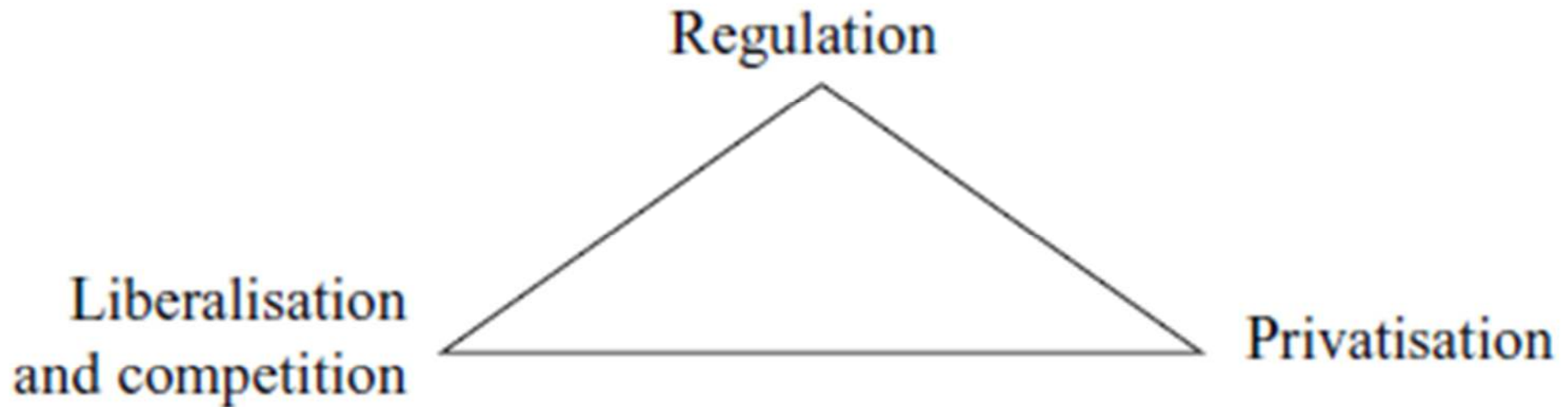
Telecommunication Sector Reforms

- **Regulation:**

- ❖ This refers to the intervention by governments in markets in order to achieve certain outcomes in those markets.
- ❖ It can also be defined as a form of market intervention that aims to stimulate behaviors that would not by themselves emerge.
- ❖ It is undertaken to encourage desirable outcomes or to remedy proven problems.
- The regulatory process includes
 - ❖ developing, agreeing, setting, evolving and enforcing rules of conduct and engagement.



Telecommunication Sector Reforms



Assignment (Group)

- Is it good to privatize public assets?



Overview of Telecommunications Regulation

- ❖ Regulatory objectives
- ❖ Regulatory organisations
- ❖ The regulatory process
- ❖ Principles of effective regulation



Regulatory objectives

- Regulation takes three basic forms;
- ❖ **Technical standardization**, which establishes a common and legally enforceable platform for all businesses within a sector or across the whole of business
 - e.g., the rules governing electrical power supply or company accounting.
- ❖ **Public policy regulation**, which seeks to achieve objectives because they are believed to be in the public interest
 - e.g., legislation which controls the supply of particular goods and services, such as alcohol, pornography and gambling.



Regulatory objectives

- ❖ **Competition regulation**, which seeks to create and maintain fair competitive relationships between businesses, and between businesses and consumers.



Principal Focus of Regulation in the Telecoms Sector

- Regulation in the telecom sector may take all three basic forms.
- ❖ However, the principal focus for contemporary regulation of the telecommunications services industry is the **creation, nourishment and maintenance of competitive markets.**



Widely Accepted Regulatory Objectives in the Telecoms Sector

- ❖ Promote universal access to basic telecommunications services
- ❖ Foster competitive markets to promote:
 - efficient supply of telecommunications services
 - good quality of service
 - advanced services
 - efficient prices



Widely Accepted Regulatory Objectives in the Telecoms Sector

- ❖ Where competitive markets do not exist or fail, prevent abuses of market power such as excessive pricing and anti-competitive behavior by dominant firms.
- ❖ Create a favorable climate to promote investment to expand telecommunications networks
- ❖ Promote public confidence in telecommunications markets through transparent regulatory and licensing processes
- ❖ Protect consumer rights, including privacy rights



Widely Accepted Regulatory Objectives in the Telecoms Sector

- ❖ Promote increased telecommunications connectivity for all users through efficient interconnection arrangements
- ❖ Optimize use of scarce resources, such as the radio spectrum, numbers and rights of way



Policy and Regulation

- ❖ Policy can be defined as,
 - a deliberate plan of action to guide decisions and achieve rational outcomes
 - may apply to government, private sector organizations and groups, and individuals.
- ❖ Policies are generally put into practice through legislation which provides a framework for regulating the conduct of citizens, businesses and other actors in society



Policy and Regulation

- While law can compel or prohibit behaviors, policy merely guides actions toward those that are most likely to achieve a desired outcome
- ❖ Regulation focuses on translating the objectives of policy decisions into practical measures.
- ❖ Whereas policy making is primarily a matter of political decision making, regulation is much more technocratic, concerned with identifying the most effective ways of achieving the objectives which policy makers have agreed.



Policy and Regulation

- ❖ Regulation is done primarily through rules and regulations, and by decisions about specific circumstances which arise in which those rules and regulations must be interpreted.
 - Rules and regulations may be implemented:
 - ex ante – before certain issues arise
 - ex post – after some issues have already occurred



The Role of National Government Authorities

- ❖ Formerly, in many countries, a single ministry or other government administrative unit acted as policy makers, policy implementers and operators of the telephone service.
- ❖ The most common institutional model used in developed market economies around the world today is shown in the following table;



The Role of National Government Authorities

Task	Entity/entities
Policy Development	Government Ministry or Executive Branch
Regulation	Separate Regulatory Authority
Network Operations/ Service Provision	Privately or commercially operated



International Agencies

- ❖ Apart from government and other agencies of the state, some international bodies also play important roles in telecommunications policy and regulation.
- ❖ Examples include the following;
 - International Telecommunication Union (ITU)
 - World Trade Organization (WTO)
 - African Telecommunications Union (ATU)
 - West African Telecommunications Regulators Association (WATRA),
etc.



International Telecommunication Union (ITU)

- ❖ ITU is a specialized agency of the United Nations (UN) responsible for issues that concern information and communication technologies.
- ❖ The mission of ITU include
 - coordinating the **shared** global use of the radio spectrum
 - promoting international cooperation in assigning satellite orbits
 - working to improve telecommunication infrastructure in the developing world
 - assisting in the development and coordination of worldwide technical standards



International Telecommunication Union (ITU)

- ❖ The ITU is active in areas including the following;
 - broadband Internet, latest-generation wireless technologies, aeronautical and maritime navigation, radio astronomy, satellite-based meteorology, convergence in fixed-mobile phone, Internet access, data, voice, TV broadcasting, and next-generation networks.
- ❖ ITU also organizes worldwide and regional exhibitions and forums, such as “ITU Telecom World”, ITU – WRC Conferences
- ❖ Such platforms bring together representatives of governments and the telecommunications and ICT industry to exchange ideas, knowledge and technology.



ITU Sectors

- ❖ The ITU comprises three sectors, each managing a different aspect of the matters handled by the Union
- ❖ The 3 sectors are as follows;
 - ✓ **Radio-communication (ITU-R):** This sector manages the international radio frequency spectrum and satellite orbit resources.
 - ✓ **Standardization (ITU-T):** This sector standardizes global telecommunications (except for radio).
 - ✓ **Development (ITU-D):** This sector helps spread equitable, sustainable and affordable access to information and communication technologies (ICT).



World Trade Organization

- ❖ The World Trade Organization (WTO) is an inter-governmental organization which regulates international trade.
- ❖ The organization deals with regulation of trade between participating countries
 - by providing a framework for negotiating trade agreements and a dispute resolution process aimed at enforcing participants' adherence to WTO agreements, which are signed by representatives of member governments and ratified by their parliaments.



World Trade Organization

- ❖ The World Trade Organization's Agreement on Basic Telecommunications Services (the BTA) establishes rules for competition in international and national telecoms and promotes open (international) investment in telecommunications markets.
 - The Agreement includes a Reference Paper on Regulation which sets out basic rules or guidelines for telecoms regulation.



The WTO Reference Paper on Regulation

- ❖ The rules or guidelines contained in the Reference Paper represent global consensus on aspects of telecoms regulation.
- ❖ They require adherence to six basic principles;
 - There should be a regulatory body independent of operating companies.
 - There should be safeguards to prevent anti-competitive behavior.



The WTO Reference Paper on Regulation

- Licensing criteria should be publicly available and transparent.
- Regulation should provide for interconnection between major operators.
- It should apply any **universal access obligations** which are agreed in a way which is transparent and neutral between operators.
- It should allocate **scarce resources** (such as numbers and spectrum) fairly between operators.



Regulatory governance

- ❖ The WTO Reference Paper specifies that telecoms regulators should be “independent” of operating companies (but not necessarily of governments).
- ❖ Stakeholders in the industry think it is also better for regulators to be as separate as possible from ministries and other official government agencies. The reasons are as follows;
 - In many countries, the government still owns part or all of at least one telecommunications company.



Regulatory governance

- Consistent regulation is needed in order to achieve the confidence of investors, operators and consumers.
 - **Ministries** are less knowledgeable about the market
- Their involvement in regulatory decisions can undermine the authority of regulators and therefore confidence in markets.



Regulatory governance

- ❖ In most countries nowadays, regulatory authorities are made up of commissions.
- ❖ The commissions consist of a number of people who are usually non-executives appointed by the government.
 - Some countries include representatives of consumer groups within regulatory commissions.
 - Others also have established consumer committees to provide advice and monitor the impact of decisions.



Regulatory governance

- Many, however, do not have any direct consumer or civil society involvement.
- ❖ Commissions make decisions on the basis of recommendations put to them by the chief executive and permanent staff of the regulatory agency.



The Regulatory Process

- ❖ Regulators employ a variety of regulatory procedures.
- ❖ Depending on the legal framework, they may issue different types of regulatory instruments
 - e.g., regulations, decisions, orders, decrees, rules, policies, notices, resolutions
- ❖ In general, the effect of these instruments is to make decisions that implement **regulatory policies**, **resolve disputes**, or **deal with other matters** within the regulators' mandate.



The Regulatory Process

❖ The principles of good regulatory decision-making include:

➤ Transparency

➤ Objectivity

➤ Professionalism

➤ Efficiency

➤ Independence



The Regulatory Process

- ❖ Two “fundamental rules” of **procedural fairness** in common law countries are worth noting.
 - Provide all interested parties with an opportunity to comment or otherwise make their case, before making a decision that affects them. This rule is sometimes expressed by means of the Latin maxim *audi alteram partem* or “hear the other side”.



The Regulatory Process

- “Do not be a judge in your own case”. This rule is based on another Latin legal maxim: *nemo judex in sua causa debet esse*. The rule has been interpreted to mean that regulators should avoid bias as well as the perception of bias. In the words of the jurisprudence: “justice must not only be done, it must be seen to be done”.



The Regulatory Process

- ❖ While these common law rules are not mandatory and do not cover all the bases of good decision making, they will promote credible and impartial good decision-making.
- ❖ Various other rules and principles for good regulatory decision-making have been promulgated by different regulators.
- **Discussion (Group):** Is the telecommunications sector regulator in your country really independent? Why?



Assignment: Group

- ❖ Does the legislation and any subsequent amendments establishing the telecommunications regulator in a country of your choice (e.g. NCA in Ghana or NCC in Nigeria) address all the principles in the WTO reference paper?
 - If some aspects of the WTO reference paper are not addressed by the legislation, give recommendations for addressing them.

